INDEPENDENT ANALYSIS

Systematic Underestimation in Defense Cost and Schedule Estimates

Evidence from GAO's 2024 and 2025 Weapon Systems Annual Assessments

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AGENDA

Presentation Overview

01	**Background**	The scale of the problem
02	**Methodology**	Data and approach
03	**Findings**	Schedule, cost, and reference class
04	**Recommendations**	Practical changes and limitations
05	**Key Takeaways**	Summary and discussion

BACKGROUND

Major Program Estimates Grew $49.3 Billion in One Year

$49.3B
combined cost growth across 30 major programs (106 reviewed)

73%
of the total from one program: Sentinel

+18 mo
added to time to initial capability; average now nearly 12 years

Are these errors random, or do they consistently lean in one direction?

Source: GAO-25-107569, Weapon Systems Annual Assessment (June 2025)

THEORETICAL BASIS

Research Identifies Two Causes of Underestimation

258 transport projects **9 in 10** over budget **+28%** average overrun

Optimism bias

Planners rely on the best-case scenario for their own project (the "inside view").

Strategic misrepresentation

Low estimates help projects secure approval and funding.

Both push estimates low.

Remedy for both: benchmark against the **actual** outcomes of comparable projects (the "outside view"), which Flyvbjerg turned into **reference class forecasting**.

Flyvbjerg, Holm & Buhl (2002); Kahneman & Tversky (1979); Kahneman & Lovallo (1993); Flyvbjerg (2006)

METHODOLOGY

31 Programs Across Two GAO Assessments

01

Collect

GAO program tables, 2024 and 2025

02

Filter

31 programs; 22 with schedule data

03

Adjust

FY2025 dollars; checked on 5 programs

04

Measure

Current ÷ first (1.3 = 30% growth)

Kept

Major defense acquisition programs (MDAPs) with a first full estimate to compare against

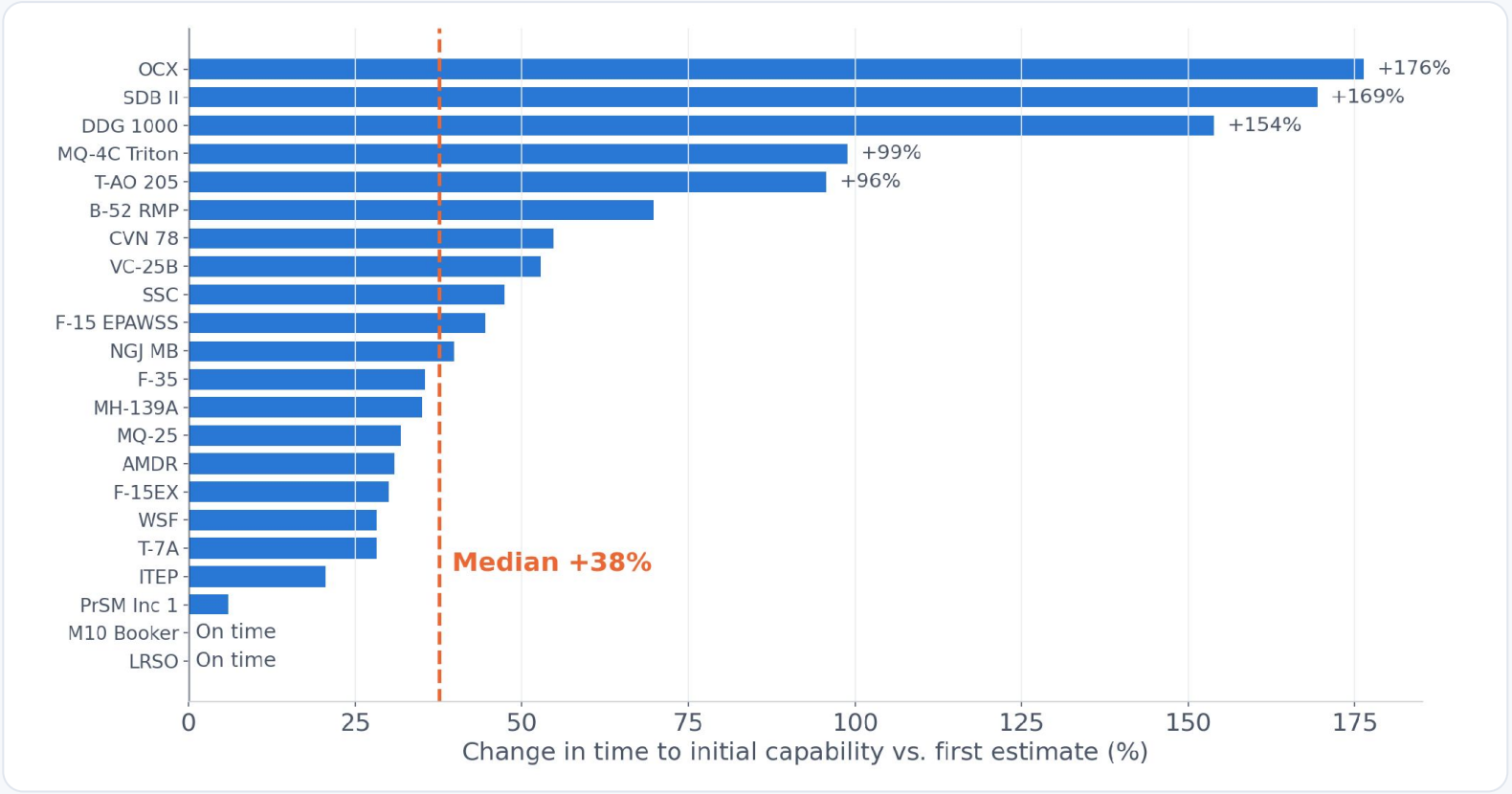
Filtered out

- Middle Tier programs (no first full estimate)
- Increments and future acquisitions (no comparable data)
- Programs with no estimate yet, or whose first estimate is still their current one (no history to measure)

Analysis in Python: pandas, NumPy, Matplotlib

FINDING 1 · SCHEDULE

20 of 22 Programs Have Slipped; None Is Ahead of Schedule



0

programs ahead of schedule

+38%

median schedule growth

Sign test: 20 of 20 changes were late. Odds by chance: about 1 in a million.

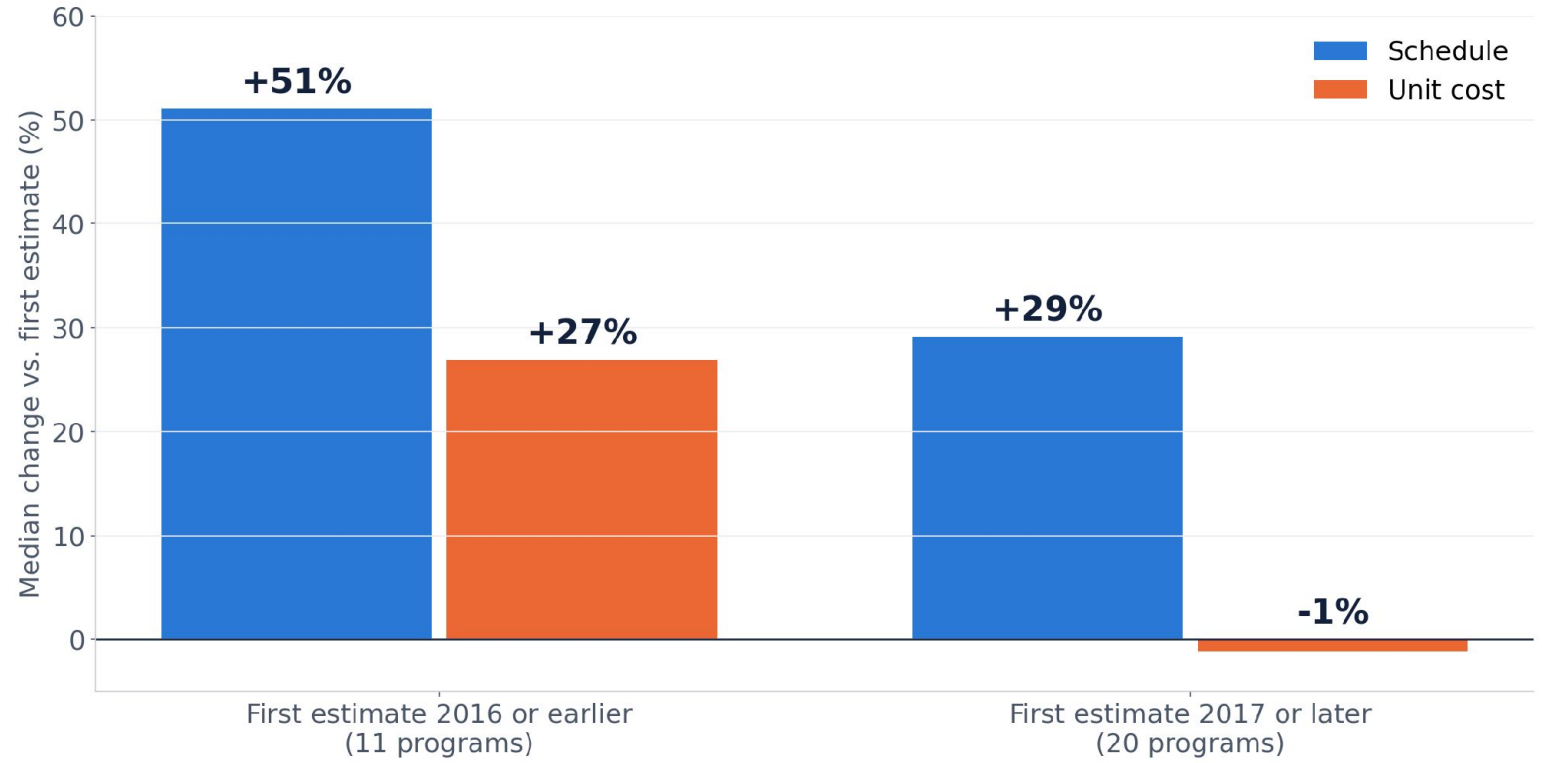
2024→2025: 12 slipped further; none recovered.

e.g., B-52 RMP: 73 → 107 months in one year.

Source: GAO-24-106831, GAO-25-107569 (MDAPs only). Analysis by LS.

FINDING 2 · COST

Schedule Growth Appears to Precede Cost Growth



+27%

unit cost, programs estimated 2016 or earlier

−1%

unit cost for newer programs, yet schedule is already +29%

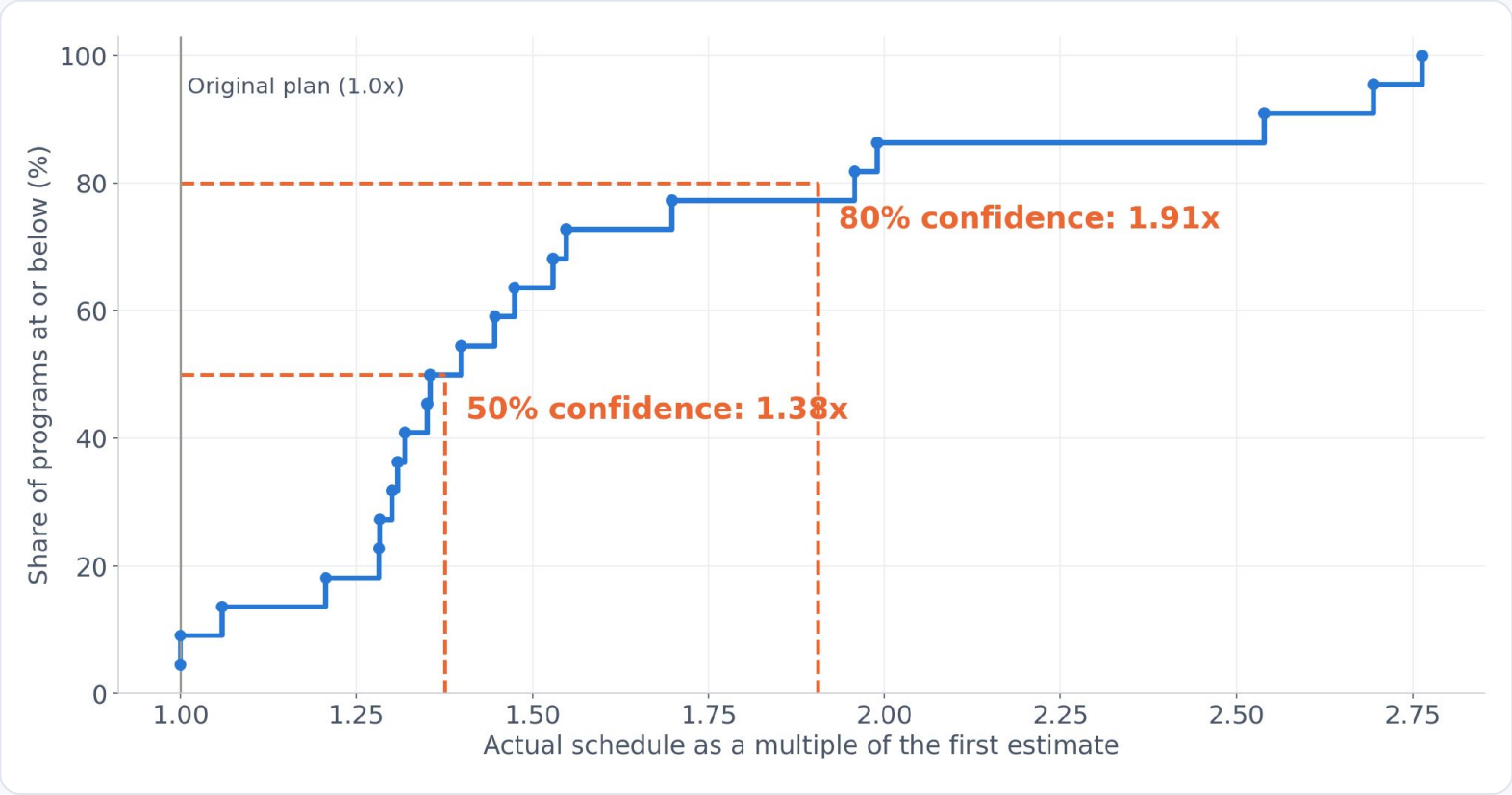
30 programs with unit cost data (MGUE has none): 16 over, 14 under.

Together, cost looks balanced, but that mixes older programs (cost already grew) with newer ones (not yet). Separated by age, the pattern is clear.

Source: GAO-24-106831, GAO-25-107569 (MDAPs only). Analysis by LS.

FINDING 3 · REFERENCE CLASS

80% Schedule Confidence Requires About 1.9 Times the Original Plan



1.38x
to hit the date half the time

1.91x
to hit it 80% of the time

Bootstrap range: 1.5x to 2.7x

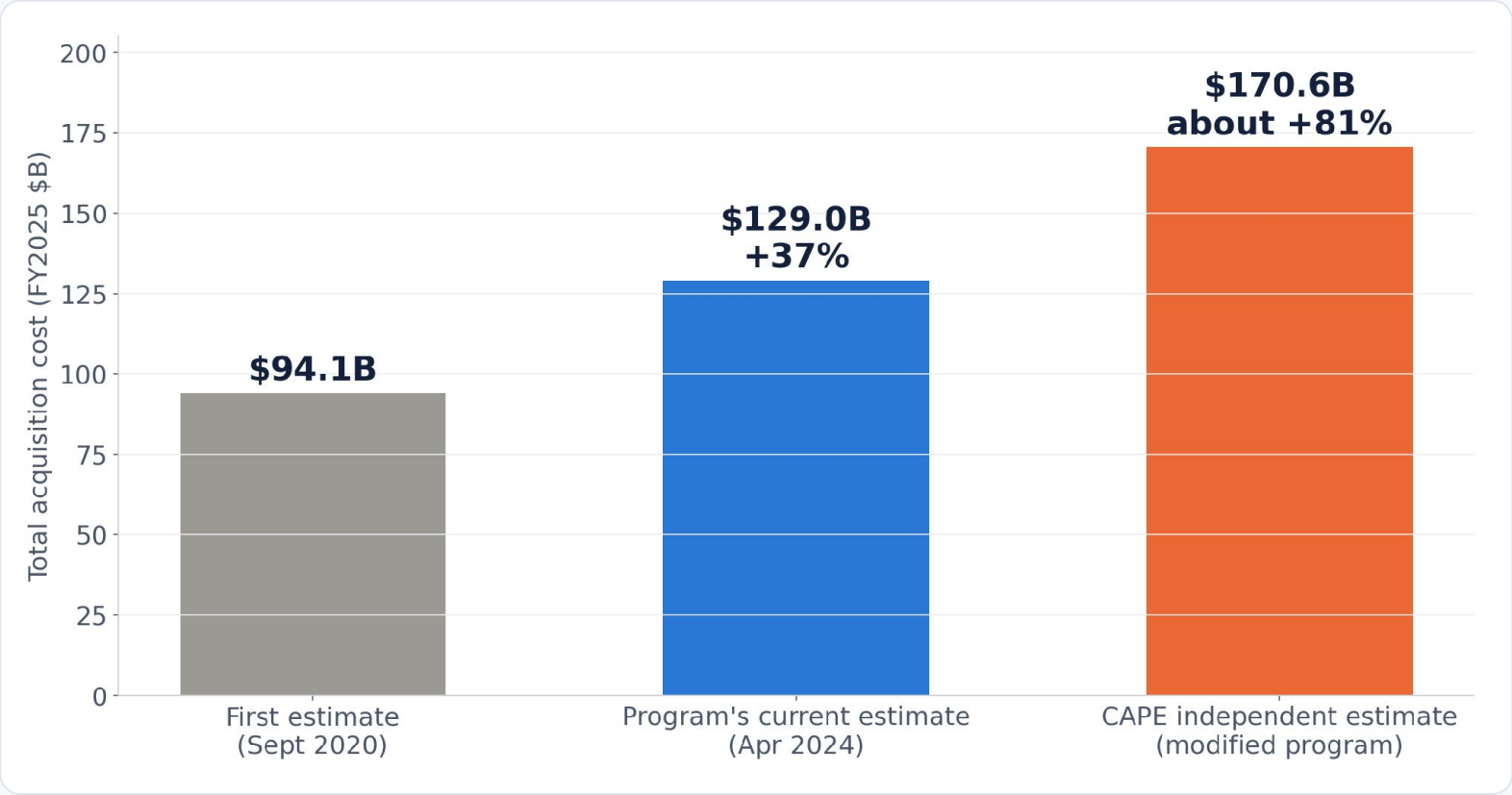
100 month plan → about 191 months

5 of 22 went beyond 1.9x (e.g., OCX, DDG 1000)

Source: GAO-24-106831, GAO-25-107569 (MDAPs only). Analysis by LS.

CASE STUDY

Sentinel: Independent Estimate Exceeds Program Estimate by $41.6 Billion



+37%

program office estimate vs. first estimate

~+81%

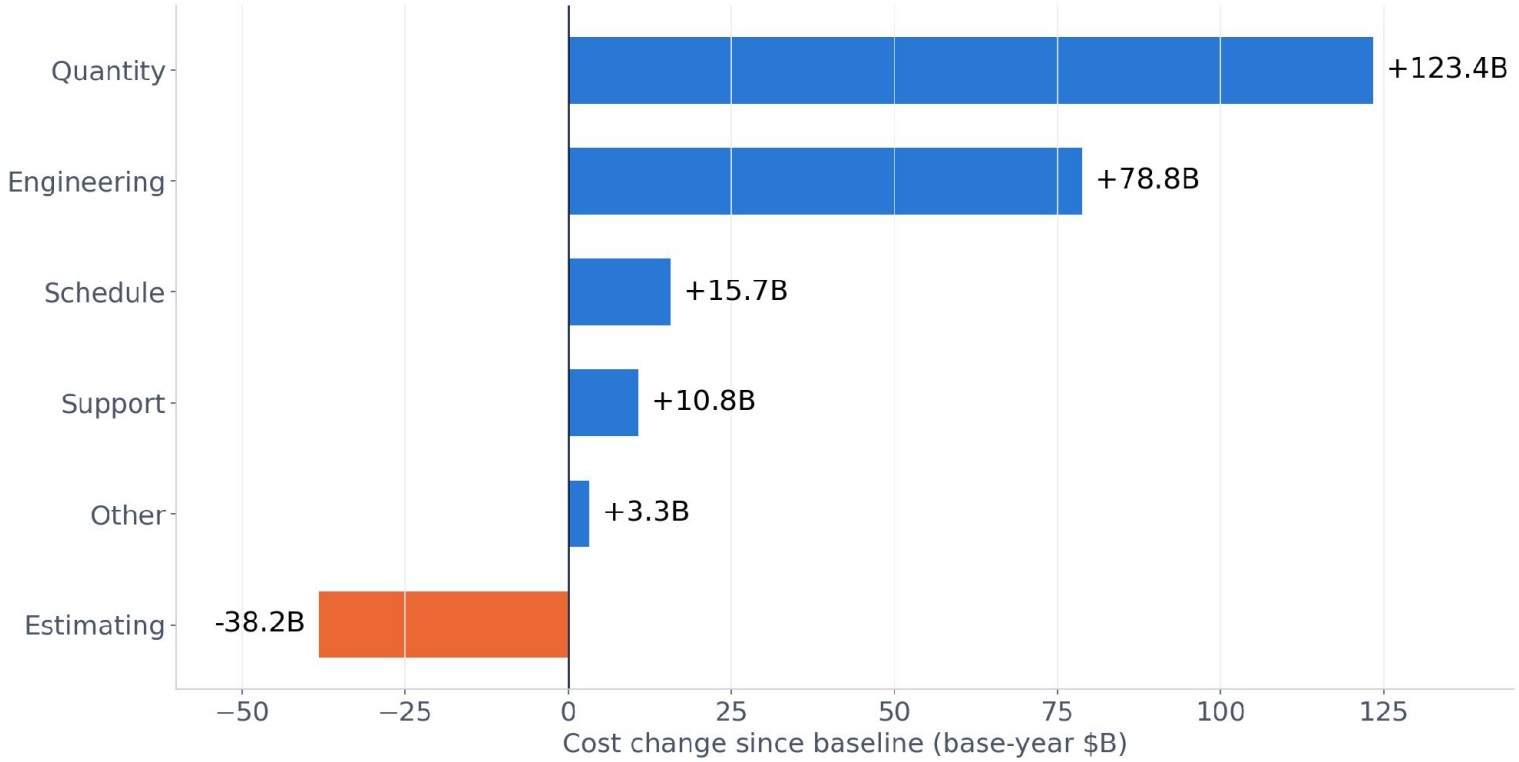
independent CAPE estimate vs. first estimate

Critical Nunn-McCurdy breach (Jan 2024): When cost grows past a set limit, DoD must review the program and report to Congress.

Source: GAO-25-107569 (June 2025). FY2025 dollars. CAPE (Cost Assessment and Program Evaluation).

ALTERNATIVE EXPLANATION

DoD Attributes Most Cost Growth to Quantity and Engineering Changes



WHAT DOD'S DATA SAYS

Buying more units	**+$123B**
Design changes	**+$79B**
Fixing the estimates	**−$38B**

So bad estimates aren't the main cause.

WHY MY FINDING STILL HOLDS

- Design changes often come from early optimism (e.g., Sentinel's immature technology)
- Buying more doesn't explain delays, yet 20 of 22 are late and none ahead

Source: DoD Selected Acquisition Report Summary Tables, Dec 2019 (Grand Total, base-year dollars). Net +$193.7B.

ALTERNATIVE EXPLANATION (CONT.)

GAO Shows Several Reported Decreases Reflect Accounting Changes, Not Savings

Program	Total cost change	What was behind it
SSBN 826	−7% in one year (2024 report)	Updated inflation assumptions; grew 6% the next year after a schedule breach
T-7A	−8% in one year	Removed its inflation contingency; initial capability ~1 year later
PrSM Inc 1	−5% in one year	Software costs moved to a different budget
FFG 62	−8% vs. first estimate	Excludes pending contractor claims; lead ship ~3 years late
DDG 1000	−34% vs. first estimate	Cut from 32 ships to 3; cost per ship about 7x

So what? Reported cost can hide real growth. Schedule is harder to hide, so it is the more reliable signal.

Source: GAO-24-106831, GAO-25-107569. Figures are total-cost changes as reported by GAO; Appendix A shows unit cost versus first estimate.

RECOMMENDATIONS

Recommendations to Strengthen Early Estimates

01

Report schedule as a range

Show 50% and 80% dates, not a single date

02

Anchor to past programs

Use a reference class alongside CAPE's independent estimate

03

Watch schedule as an early warning

Flag early slips for cost review

Builds on existing practice:

The law already requires cost estimates to discuss risk, and GAO's guides recommend risk analysis. These steps extend that to schedule and add past-program data.

Sources: 10 U.S.C. § 3225 (originally WSARA 2009); GAO Cost Estimating and Assessment Guide (GAO-20-195G); GAO Schedule Assessment Guide (GAO-16-89G).

LIMITATIONS

Limitations and Next Steps

Limits

- 31 programs; 22 with schedule data
- Current estimates, not final costs
- Cannot separate estimating error from scope changes

Next steps

- Full SAR history, by program type
- Back-test on completed programs
- Attribute growth by cause

The limits affect precision, not direction.

20 of 22 are late and none are ahead. Most programs aren't finished, so these results likely understate the problem.

CONCLUSION

Key Takeaways

01

Estimates lean low

20 of 22 programs are behind schedule; none is ahead.

02

Plan with ranges

About 1.4x the plan for 50% confidence; about 1.9x for 80%.

03

Watch schedule first

Early schedule slips are a warning sign for cost growth.

An honest range is more useful than one optimistic number.

Thank You!

Any Questions?

APPENDIX A

Program Data: Change Versus First Full Estimate

Program	Unit cost	Schedule
AARGM-ER	+6%	—
AMDR	−48%	+31%
B-52 RMP	+10%	+70%
CH-47F Block II	−10%	—
CVN 78	−1%	+55%
DDG 1000	+605%	+154%
F-15 EPAWSS	+40%	+45%
F-15EX	−3%	+30%
F-35	+79%	+35%
FFG 62	−8%	—
GPS IIIF	−12%	—

Program	Unit cost	Schedule
IRST	+6%	—
ITEP	+6%	+21%
KC-46A	−27%	—
LRSO	−7%	0%
M10 Booker	+3%	0%
MGUE Inc 1	—	—
MH-139A	+6%	+35%
MQ-25	+1%	+32%
MQ-4C Triton	+149%	+99%
NGJ MB	+9%	+40%

Program	Unit cost	Schedule
OCX	+69%	+176%
PrSM Inc 1	−2%	+6%
SDB II	−4%	+169%
Sentinel	+37%	—
SSBN 826	−1%	—
SSC	+27%	+47%
T-7A	−13%	+28%
T-AO 205	+24%	+96%
VC-25B	−2%	+53%
WSF	−5%	+28%

Unit cost = total acquisition cost ÷ quantity. “—” = not reported or TBD. Source: GAO-24-106831, GAO-25-107569.

APPENDIX B

Methodology Detail: Dataset Construction

Included

- 31 MDAPs with a first full estimate
- 22 with schedule data
- F-35 from 2024 report

Excluded

- Middle Tier programs
- Increments and future programs
- No estimate or brand-new baseline

Adjusted

- FY2024 × 1.023 = FY2025 dollars
- Checked on 5 programs in both reports
- Unit cost = total ÷ quantity

Sources: GAO-24-106831 (June 2024), GAO-25-107569 (June 2025).

APPENDIX C

References

1. U.S. Government Accountability Office. *Weapon Systems Annual Assessment*, GAO-24-106831. June 2024.
2. U.S. Government Accountability Office. *Weapon Systems Annual Assessment*, GAO-25-107569. June 2025.
3. U.S. Department of Defense. *Selected Acquisition Report (SAR) Summary Tables*. December 2019.
4. U.S. Government Accountability Office. *Cost Estimating and Assessment Guide*, GAO-20-195G. March 2020.
5. U.S. Government Accountability Office. *Schedule Assessment Guide*, GAO-16-89G. December 2015.
6. Weapon Systems Acquisition Reform Act of 2009, Pub. L. 111-23; now 10 U.S.C. § 3225 (formerly § 2334), as amended by Pub. L. 114-328 (2016).
7. Flyvbjerg, B., Holm, M. S., & Buhl, S. (2002). Underestimating costs in public works projects: Error or lie? *Journal of the American Planning Association*, 68(3).
8. Flyvbjerg, B. (2006). From Nobel Prize to project management: Getting risks right. *Project Management Journal*, 37(3).
9. Kahneman, D., & Tversky, A. (1979). Intuitive prediction: Biases and corrective procedures. *TIMS Studies in Management Science*, 12.
10. Kahneman, D., & Lovallo, D. (1993). Timid choices and bold forecasts. *Management Science*, 39(1).